

PCA: Problem Formulation and Step-by-Step Solution

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data), 6 (Reduce dimensions). See the [Course map](#).



- **Builds on:** Unsupervised learning ([Note 3](#)); Standardization ([Note 24](#)); Feature scaling ([Note 24](#)); Curse of dimensionality ([Note 46](#)); Feature extraction ([Note 47](#)); Variance ([Note 47](#)).
- **Leads to:** Ordinary least squares (closed form) ([Note 51](#)).
- **Compare with:** Correlation ([Note 21](#)).

1. Overview

Key point: PCA looks for the unit vector along which the projected data has the largest variance. The answer is the top eigenvector of the data's covariance matrix.

The previous Note built the idea of PCA geometrically: turn a line through the data, and keep the direction where the shadows spread the most. This Note turns that idea into mathematics and then into five concrete steps.

Every ML algorithm is, underneath, a mathematical problem with a goal to optimise, called its **objective function**. We first write PCA's objective function (Section 2). Then we learn the two tools its solution uses: the covariance matrix (Section 3) and eigenvectors (Section 4). Finally we run the five steps on real numbers (Sections 5 and 6).

2. The problem PCA solves

Key point: Project every point onto a unit vector u ; PCA wants the u that makes the variance of those projections as large as possible.

2.1 Projecting one point

Key point: The projection of a point x onto a unit vector u is a single number, $u^T x$.

Take data with two columns. Each point has an x and a y value, so we can treat it as a **vector**, an arrow from the origin to the point.

We want to project the point onto some line through the origin. Only the line's direction matters, not its length, so we describe it by a **unit vector** u : a vector of length 1 pointing along the line. Figure 1 shows a point x , a unit vector u , and the shadow of x on the line.

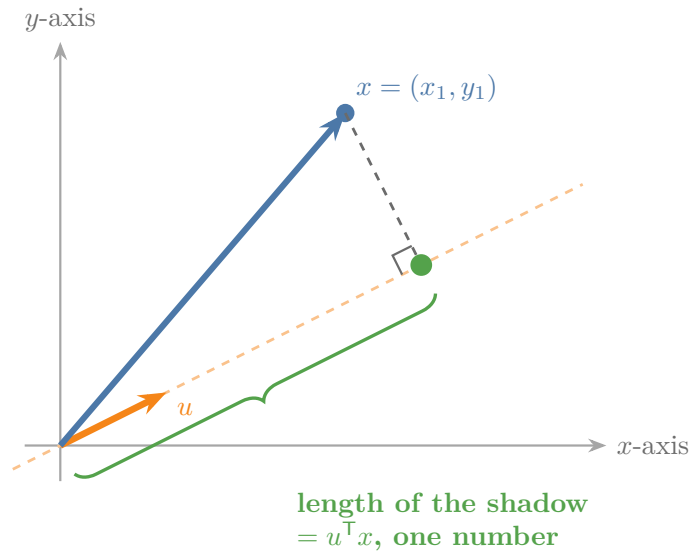


Figure 1: Projecting a point onto a unit vector

In words: the length of the shadow is the **dot product** of u and x (multiply matching components and add), divided by the length of u . Since u has length 1, the division disappears.

$$\text{shadow length} = \frac{u \cdot x}{\|u\|} = u^T x$$

With numbers: for $u = (0.894, 0.447)$, a unit vector, and the point $x = (2.4, 2.8)$:

$$u^T x = 0.894 \times 2.4 + 0.447 \times 2.8 = 2.146 + 1.252 = 3.40$$

The point is now described by one number, 3.40, its position along the line. Two columns have become one.

Extra: u^T is u written as a row instead of a column (its **transpose**). A row times a column is the dot product, so $u^T x$ is just a compact way to write $u \cdot x$.

2.2 The variance of all the projections

Key point: Projecting all n points gives n numbers; their variance tells us how much of the spread the direction u keeps.

We project every point $x_1, x_2, \dots, x_n$ the same way and get n numbers, $u^T x_1, \dots, u^T x_n$. Their mean is the projection of the mean point, $u^T \bar{x}$.

In words: the variance of the projections is the average squared distance of each projection from the projected mean.

$$\sigma^2(u) = \frac{1}{n} \sum_{i=1}^n (u^\top x_i - u^\top \bar{x})^2$$

With numbers: for the flats data of the previous Note (rooms and washrooms), this was 1.33 for u along the rooms axis and 2.61 for u at 45° .

2.3 The objective

Key point: PCA's objective: among all unit vectors, find the u with the largest $\sigma^2(u)$.

Any unit vector could be the answer, and each one gives a different variance. PCA wants the one with the most:

find u with $\|u\| = 1$ that makes $\sigma^2(u)$ as large as possible

This is PCA's objective function. Solving it needs optimisation methods covered later; here we jump to the answer. To understand the answer, we need two ideas first: covariance and eigenvectors.

Extra: The standard derivation rewrites $\sigma^2(u)$ as $u^\top C u$, where C is the covariance matrix of Section 3. Maximising $u^\top C u$ under the condition $u^\top u = 1$ (with a method called Lagrange multipliers) gives exactly $Cu = \lambda u$: the eigenvector equation of Section 4.

3. Covariance and the covariance matrix

Key point: Variance describes the spread along one column. Covariance describes how two columns move together. The covariance matrix holds both for every column and every pair.

3.1 What variance cannot see

Key point: Two datasets can have identical variances in every column and still point in opposite directions.

Variance is computed one column at a time. It says how spread out x is and how spread out y is, but nothing about how x and y relate. Figure 2 shows two tiny datasets.

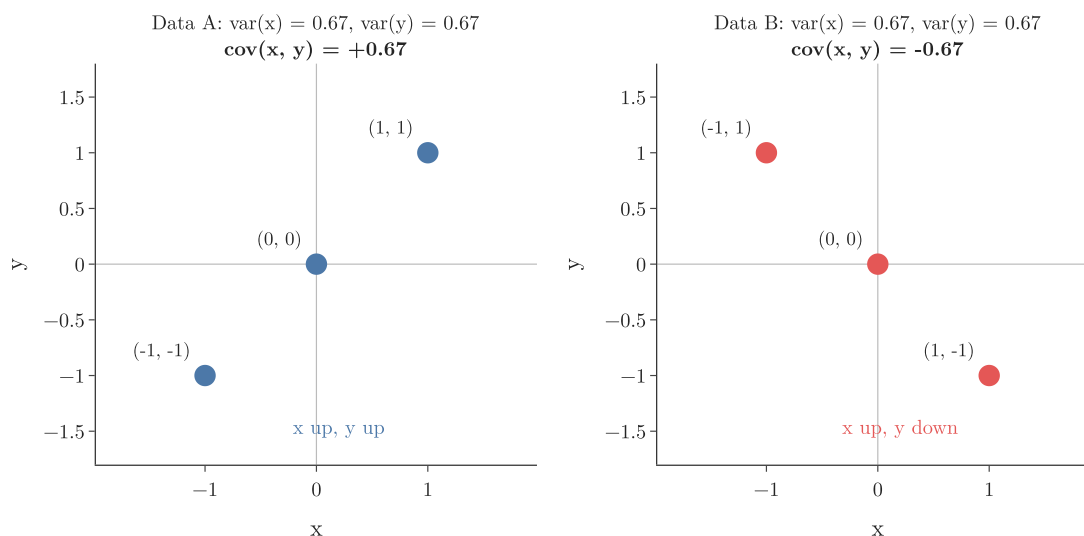


Figure 2: Same variances, opposite covariance

- **Data A:** $(-1, -1), (0, 0), (1, 1)$.
- **Data B:** $(-1, 1), (0, 0), (1, -1)$.

In both, x has variance 0.67 and y has variance 0.67. Yet in A the points rise from left to right, and in B they fall. Variance alone cannot tell them apart.

3.2 Covariance

Key point: Covariance is positive when two columns rise together and negative when one rises as the other falls.

In words: for each point, multiply its distance from the mean in x by its distance from the mean in y , then average these products.

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

With numbers: in both datasets the means are 0.

$$\text{Data A: } \frac{(-1)(-1) + (0)(0) + (1)(1)}{3} = \frac{2}{3} \approx +0.67$$

$$\text{Data B: } \frac{(-1)(1) + (0)(0) + (1)(-1)}{3} = -\frac{2}{3} \approx -0.67$$

The sign tells the direction of the relationship: positive in A, where x and y go up together, negative in B, where one goes up as the other goes down.

Extra: Correlation (from the Note on understanding data) is covariance divided by the two standard deviations. That squeezes it into the range -1 to $+1$. Covariance has no fixed range: its size depends on the units of the columns.

3.3 The covariance matrix

Key point: The covariance matrix puts each column's variance on the diagonal and each pair's covariance off the diagonal. It describes both the spread and the orientation of the data.

For data with columns x , y and z , the **covariance matrix** is the 3×3 table in Figure 3. With d columns, it is $d \times d$.

	x	y	z	
x	$\text{var}(x)$	$\text{cov}(x, y)$	$\text{cov}(x, z)$	diagonal: the variance of each column (the spread) off the diagonal: the covariance of each pair (the orientation); same colour = same value, so the matrix is symmetric
y	$\text{cov}(y, x)$	$\text{var}(y)$	$\text{cov}(y, z)$	
z	$\text{cov}(z, x)$	$\text{cov}(z, y)$	$\text{var}(z)$	

Figure 3: The covariance matrix of three columns

- **The diagonal** holds the variance of each column. The covariance of a column with itself is its variance: $\text{cov}(x, x) = \text{var}(x)$.
- **Off the diagonal** sit the covariances of each pair of columns.
- **It is symmetric:** $\text{cov}(x, y) = \text{cov}(y, x)$, so the part above the diagonal mirrors the part below.

This makes the covariance matrix a complete summary for PCA. The diagonal gives the spread along each axis; the off-diagonal entries give the orientation of the cloud.

For the flats data (rooms and washrooms):

$$C = \begin{pmatrix} 1.33 & 1.28 \\ 1.28 & 1.33 \end{pmatrix}$$

Both columns have variance 1.33, and their large positive covariance, 1.28, says they rise together.

4. Eigenvectors and eigenvalues

Key point: A matrix moves vectors around. Its eigenvectors are the special vectors it only stretches or shrinks, without turning them.

4.1 A matrix is a transformation

Key point: Multiplying by a matrix moves every point of the plane: it can rotate, stretch, squash or shear the whole grid.

Think of a matrix as an action on the whole plane. Every point, seen as a vector, is multiplied by the matrix and lands somewhere new. Straight grid lines stay straight and evenly spaced, but the grid can be turned, stretched or slanted. Such a change is called a **linear transformation**.

The **identity matrix** $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ leaves every vector where it was. Any other matrix moves vectors.

4.2 The vectors that do not turn

Key point: An eigenvector keeps its direction under the transformation; its eigenvalue is how much it is stretched.

Figure 4 applies the matrix $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ to the whole plane and follows three vectors.

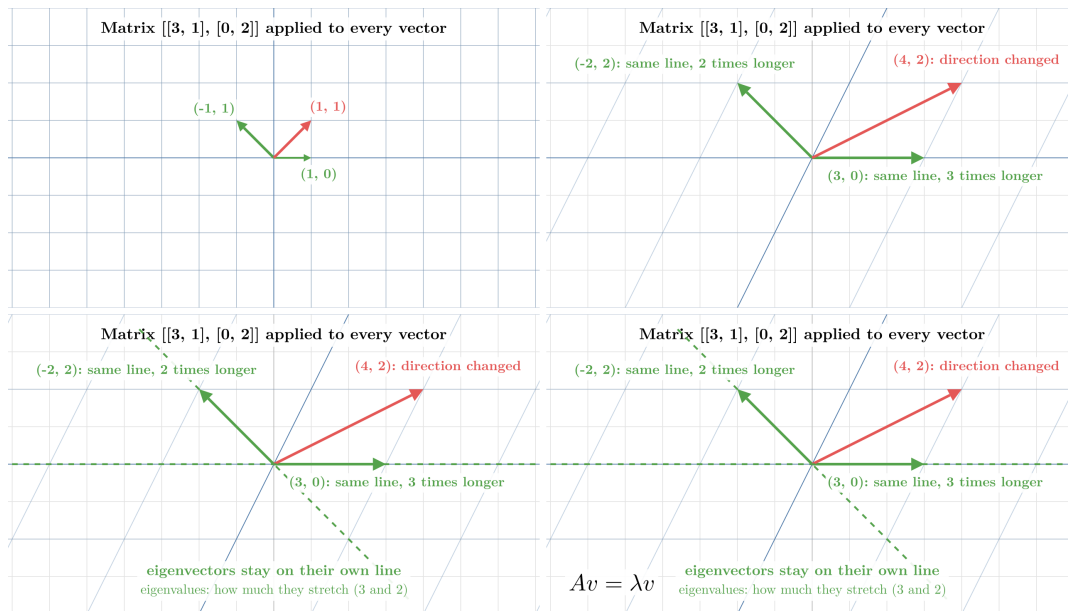


Figure 4: Applying a matrix: most vectors turn, eigenvectors only stretch

1. $(1, 1)$ (red) becomes $(4, 2)$: its direction changes.

2. $(1, 0)$ (green) becomes $(3, 0)$: same line, 3 times longer.
3. $(-1, 1)$ (green) becomes $(-2, 2)$: same line, 2 times longer.

Vectors that stay on their own line are the **eigenvectors** of the matrix. The factor by which each one is stretched is its **eigenvalue**: 3 for $(1, 0)$ and 2 for $(-1, 1)$. In two dimensions there are two eigenvector directions, in three dimensions three, and so on.

In words: applying the matrix to an eigenvector is the same as multiplying it by a plain number, its eigenvalue.

$$Av = \lambda v$$

With numbers, for $v = (-1, 1)$:

$$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \times (-1) + 1 \times 1 \\ 0 \times (-1) + 2 \times 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

so $\lambda = 2$. An eigenvalue can also be negative (the vector flips to point the other way) or between 0 and 1 (it shrinks).

4.3 The eigenvectors of the covariance matrix

Key point: The eigenvector of the covariance matrix with the largest eigenvalue points along the greatest spread of the data. That eigenvector is PC1, and its eigenvalue is the variance along it.

This is the result that solves PCA's objective from Section 2. Use the covariance matrix as the transformation and find its eigenvectors:

- The eigenvector with the **largest eigenvalue** points in the direction of greatest variance. It is the first principal component.
- Its **eigenvalue** equals the variance of the data projected onto it.
- The next eigenvector (next largest eigenvalue) is PC2, at right angles to PC1, and so on.

Figure 5 checks this on the flats data.

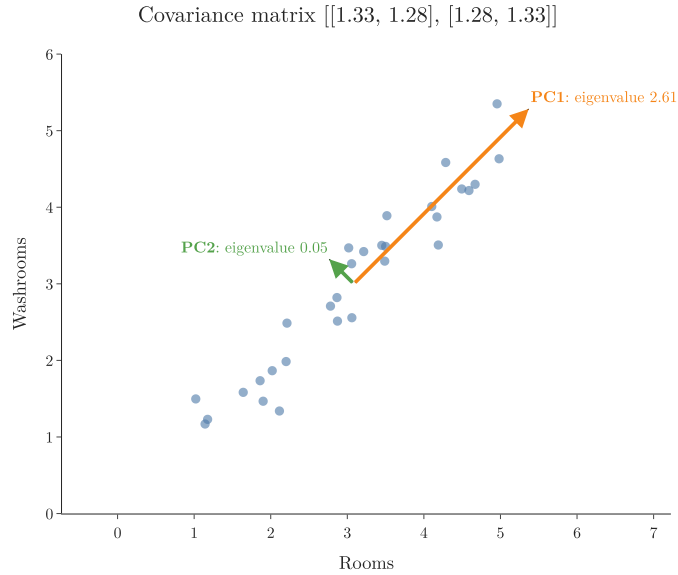


Figure 5: The eigenvectors of the covariance matrix of the flats data

The covariance matrix $\begin{pmatrix} 1.33 & 1.28 \\ 1.28 & 1.33 \end{pmatrix}$ has eigenvectors $(0.707, 0.707)$ and $(-0.707, 0.707)$, with eigenvalues 2.61 and 0.05.

The first eigenvector points at 45° , exactly the direction the previous Note found by turning a line and measuring. Its eigenvalue, 2.61, is the variance we measured there. The second has eigenvalue 0.05, the small spread left at right angles.

So instead of trying every angle, PCA computes the eigenvectors of one matrix. This works the same way with 3, 10 or 1,000 columns.

Extra: A covariance matrix is symmetric, which guarantees that its eigenvectors are at right angles to each other and that all its eigenvalues are 0 or positive. That is why the principal components form a proper new set of axes.

5. PCA in five steps

Key point: Centre the data, build the covariance matrix, find its eigenvectors, keep the top k , and project the data onto them.

Figure 6 summarises the whole method, with the shapes for data of 40 rows and 3 columns reduced to 2.

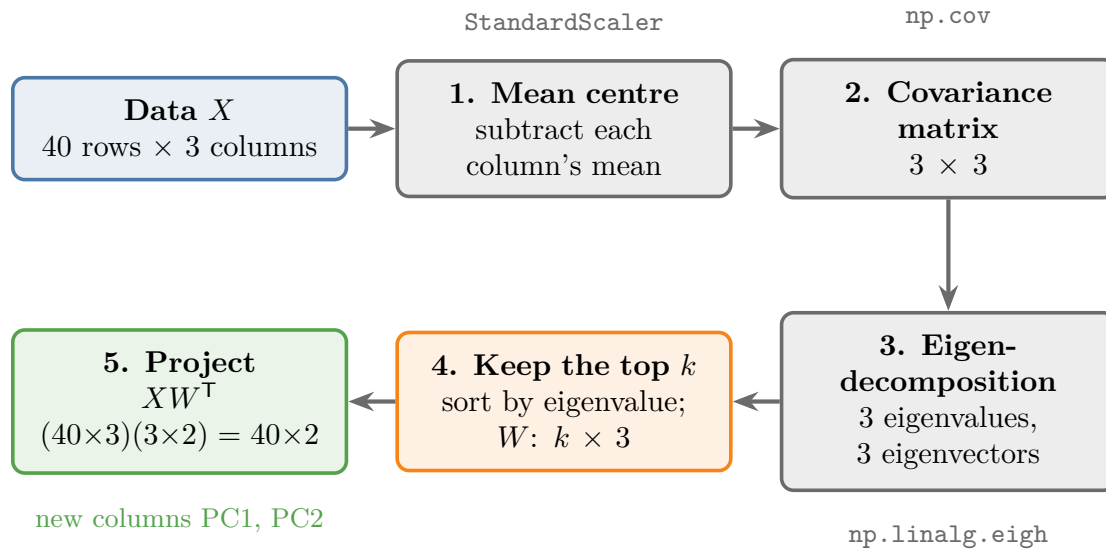


Figure 6: PCA step by step

1. **Mean centring.** Subtract each column's mean, so the data is centred on the origin.
2. **Covariance matrix.** Compute it from the centred data: 3×3 for 3 columns.
3. **Eigen-decomposition.** Find the eigenvalues and eigenvectors of the covariance matrix: 3 of each for 3 columns.
4. **Keep the top k .** Sort the eigenvectors by eigenvalue, largest first. Keep the first k as the rows of a matrix W ($k \times 3$). Choosing $k = 2$ goes from 3D to 2D; $k = 1$ goes to 1D.
5. **Project.** Multiply the data by W^T .

5.1 The projection step

Key point: Projecting all points at once is one matrix multiplication: XW^T .

In words: each point's new coordinates are its dot products with each kept eigenvector, as in Section 2.1. Doing this for all points at once is a matrix product.

$$Z = XW^T$$

With shapes: X has 40 rows and 3 columns; W^T has 3 rows and 2 columns. The inner sizes (3 and 3) match, and the result Z has 40 rows and 2 columns: the new columns PC1 and PC2.

For 1D instead, W has one row, W^T is 3×1 and Z is 40×1 . The output column, if there is one, is copied across unchanged: PCA only transforms the inputs.

Extra: Mean centring is sometimes called optional, but in practice it is needed. `np.cov` centres the data internally, so the covariance matrix and the principal components are the same either way. The projection XW^T , though, only gives centred coordinates if X

itself was centred. scikit-learn's PCA always centres for us.

6. PCA by hand in Python

Key point: The five steps take a few lines of NumPy, and the result matches scikit-learn's PCA.

The example data has 40 points with 3 columns, in two classes of 20. Each class is a cloud of points drawn at random around its own centre: (0, 0, 0) for one class and (1, 1, 1) for the other.

Python: Steps 1 to 3.

```
from sklearn.preprocessing import StandardScaler

# step 1: centre (StandardScaler also scales each column)
X = StandardScaler().fit_transform(df[cols])
# step 2: covariance matrix; np.cov wants columns as rows
C = np.cov(X.T)
# step 3: eigh is made for symmetric matrices
values, vectors = np.linalg.eigh(C)
```

The covariance matrix is

$$C = \begin{pmatrix} 1.026 & 0.205 & 0.080 \\ 0.205 & 1.026 & 0.198 \\ 0.080 & 0.198 & 1.026 \end{pmatrix}$$

All three columns have about the same variance, and the covariances are small and positive. The eigenvalues are 1.354, 0.946 and 0.778.

Python: Steps 4 and 5.

```
# sort largest eigenvalue first
order = np.argsort(values)[::-1]
values, vectors = values[order], vectors[:, order]
# eigenvectors are the COLUMNS: take the first 2 columns
W = vectors[:, :2].T          # shape (2, 3)
Z = X @ W.T                  # (40, 3) @ (3, 2) = (40, 2)
```

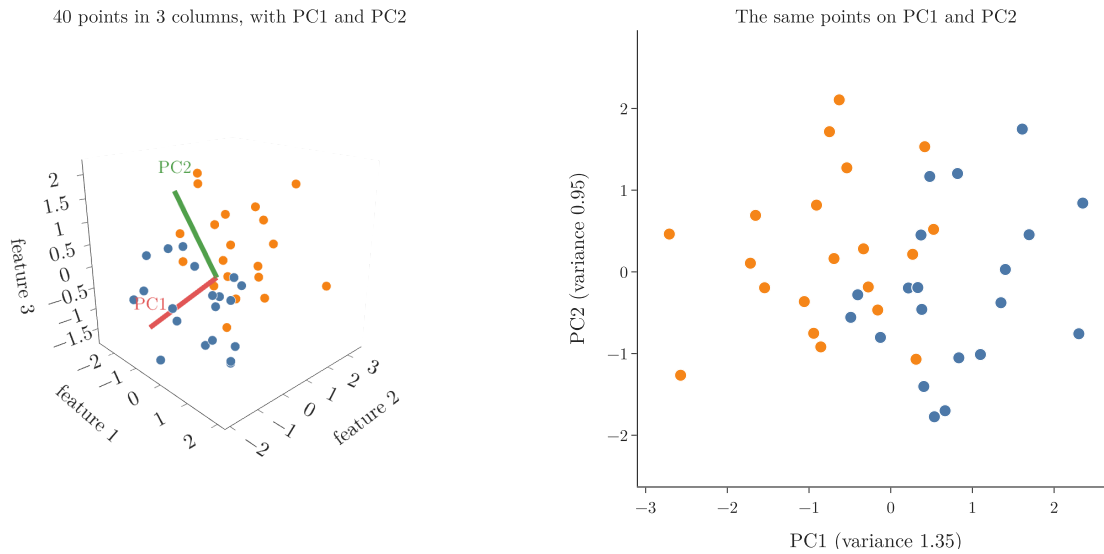


Figure 7: The 40 points in 3D with PC1 and PC2, and after projection

Figure 7 shows the result. On the left, the 3D data with the two principal components drawn through it. On the right, the same 40 points described by just PC1 and PC2.

PC1 holds 44% of the total variance and PC2 31%, so the 2D picture keeps 75% of the spread. The two classes stay mostly apart along PC1.

Extra: A frequent bug, also found in older code for this example: `np.linalg.eig` and `np.linalg.eigh` return the eigenvectors as the **columns** of the result, not its rows. Writing `vectors[0:2]` takes two rows, which are not principal components; along them the data's variance is 1.01 and 1.11 instead of 1.35 and 0.95. `eig` also does not sort its output by eigenvalue. Always sort, and take `vectors[:, :k]`.

Python: The same result from scikit-learn.
`from sklearn.decomposition import PCA`

```
pca = PCA(n_components=2)
Z_sk = pca.fit_transform(X)
pca.explained_variance_           # [1.354, 0.946]: the eigenvalues
pca.explained_variance_ratio_    # [0.44, 0.307]: share of the variance
```

The columns of `Z_sk` equal those of `Z`, possibly with the sign flipped: an eigenvector and its negative lie on the same line.

Extra: Here the variances are computed by dividing by $n - 1$ (as `np.cov` and scikit-learn do), not by n as in the formulas above. With 40 points the difference is small, and it never changes which direction is PC1.

7. Summary

Step	What it does	Tool
1. Mean centre	moves the data's centre to the origin	<code>StandardScaler</code> (or subtract the mean)
2. Covariance matrix	spread (diagonal) and orientation (off-diagonal)	<code>np.cov</code>
3. Eigen-decomposition	directions that only stretch, and by how much	<code>np.linalg.eigh</code>
4. Keep the top k	sort by eigenvalue; the first k are PC1 to PC k	<code>np.argsort</code>
5. Project	new coordinates $Z = XW^T$	@ (matrix product)

- PCA's objective: the unit vector u that maximises the variance of the projections $u^T x_i$.
- Covariance shows how two columns move together; its sign gives the direction.
- The covariance matrix: variances on the diagonal, covariances off it, symmetric.
- Eigenvectors keep their direction under a matrix; eigenvalues are their stretch, $Av = \lambda v$.
- The covariance matrix's top eigenvector is PC1; its eigenvalue is the variance along PC1.
- Eigenvectors are the columns of NumPy's output: sort them and take `vectors[:, :k]`.

8. Key terms

Term	Meaning
Objective function	The quantity an algorithm tries to make as large or as small as possible
Vector	A point seen as an arrow from the origin, with a direction and a length
Unit vector	A vector of length 1, used to describe a direction
Dot product	Multiply matching components of two vectors and add; $u^T x$
Transpose	A matrix or vector with rows and columns swapped
Covariance	How two columns move together: positive if they rise together, negative if not
Covariance matrix	A square table of all variances (diagonal) and covariances (off-diagonal) of the columns
Linear transformation	A change of the whole plane by a matrix that keeps grid lines straight and evenly spaced
Identity matrix	The matrix that leaves every vector unchanged

Term	Meaning
Eigenvector	A vector that a matrix only stretches or shrinks, without turning it
Eigenvalue	The factor by which a matrix stretches its eigenvector
Eigen-decomposition	Finding all the eigenvalues and eigenvectors of a matrix
Explained variance	The variance along a principal component; its eigenvalue